

STA 2101: Statistics & Probability

Lecture 06: Data Description (Ch 03)

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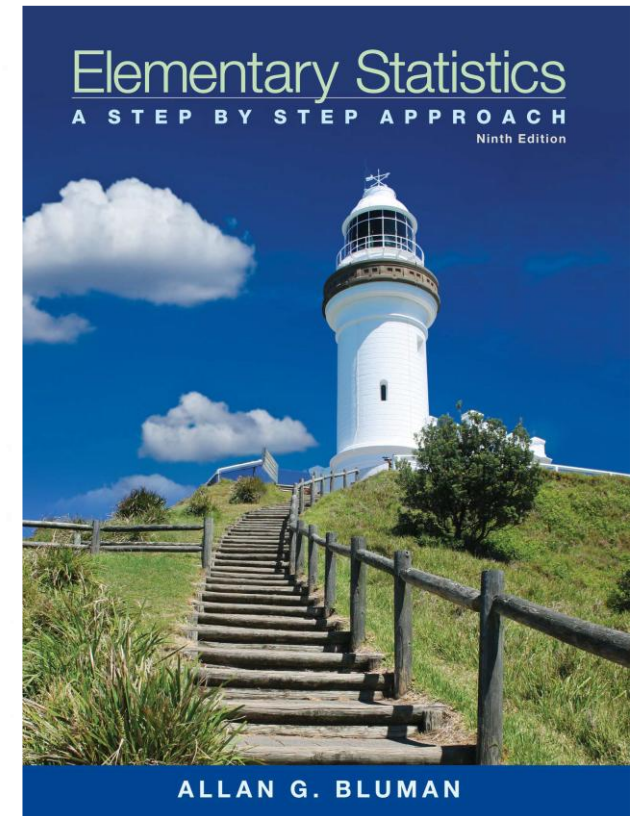
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Class Code
Sec 1 - es2uztok
Sec 5 - ayvzsvgm

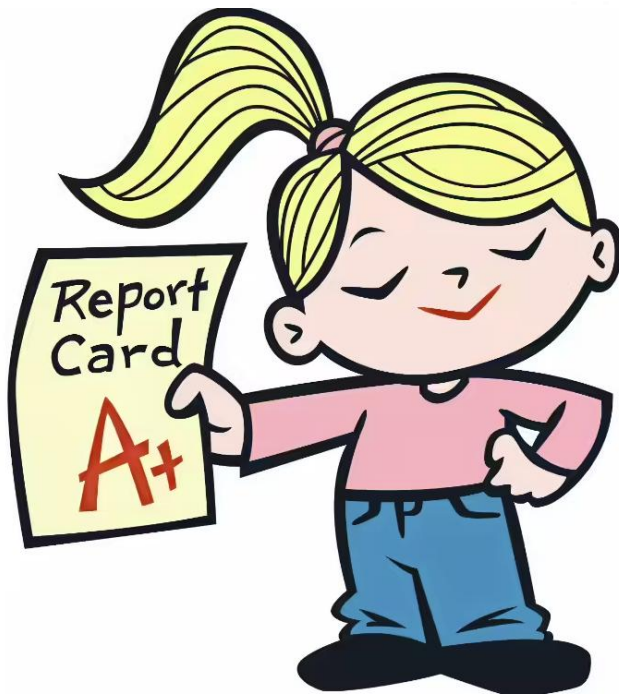
Spring 2026

Reference Book

- **Text Book(s):**
- Elementary Statistics – Allan G. Bluman
- Introduction to Probability and Statistics (14th Edition) By William Mendenhall, Robert J. Beaver, Barbara M. Beaver
- **Reference Material/Book(s):**
- S Introduction to probability and statistics for engineers and scientists, Sheldon M Ross



Mark Distribution

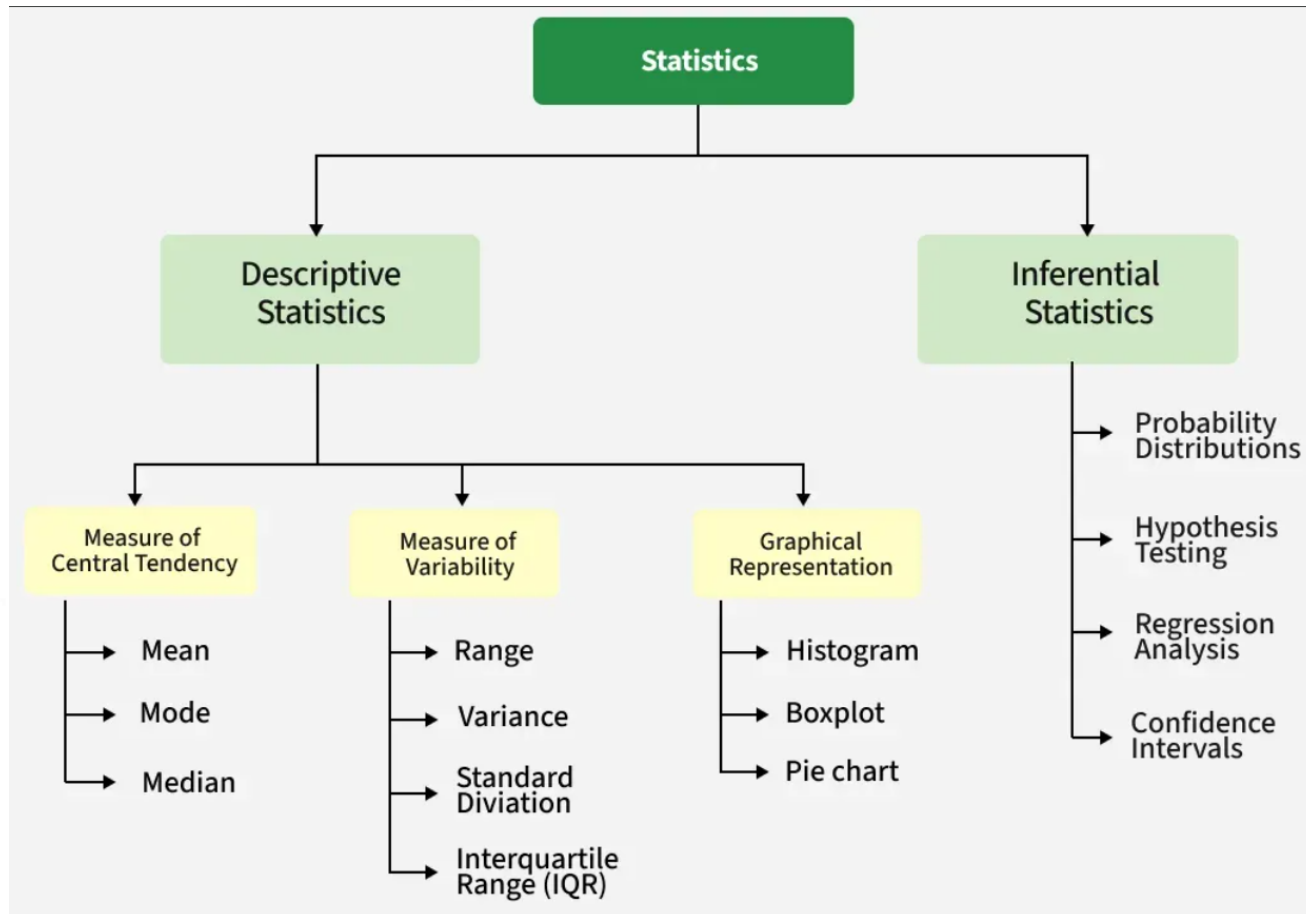


Assessments	%
Mid Term Examination	25
Final Examination	40
Attendance and Class Participation	5
CT	5
Project	25
Total	100

Today's Topics

- Measures of Central Tendency
- Mean
- Median

Let's Rewind



The Mean

The **mean** is the sum of the values, divided by the total number of values.

The **sample mean**, denoted by $\bar{X}$ (pronounced “X bar”), is calculated by using sample data. The sample mean is a statistic.

$$\bar{X} = \frac{X_1 + X_2 + X_3 + \cdots + X_n}{n} = \frac{\Sigma X}{n}$$

where n represents the total number of values in the sample.

The **population mean**, denoted by μ (pronounced “mew”), is calculated by using all the values in the population. The population mean is a parameter.

$$\mu = \frac{X_1 + X_2 + X_3 + \cdots + X_N}{N} = \frac{\Sigma X}{N}$$

where N represents the total number of values in the population.

EXAMPLE 3-1 Police Incidents

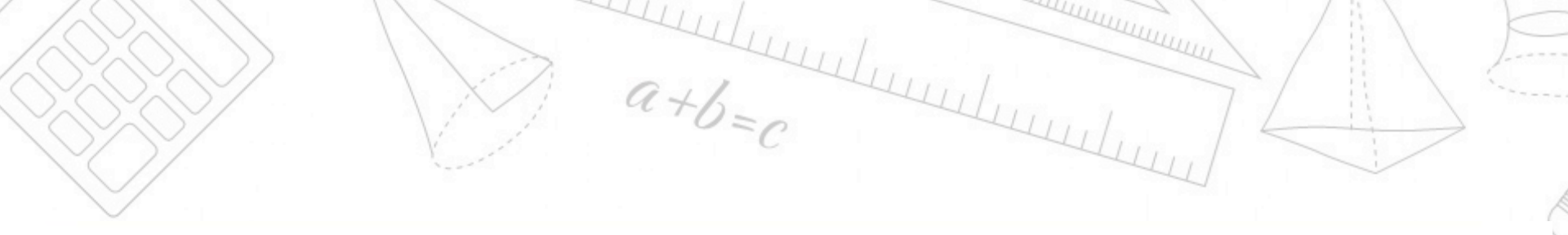
The number of calls that a local police department responded to for a sample of 9 months is shown. Find the mean. (Data were obtained by the author.)

475, 447, 440, 761, 993, 1052, 783, 671, 621

SOLUTION

$$\begin{aligned}\bar{X} &= \frac{\sum x}{n} = \frac{475 + 447 + 440 + 761 + 993 + 1052 + 783 + 671 + 621}{9} \\ &= \frac{6243}{9} \approx 693.7\end{aligned}$$

Hence, the mean number of incidents per month to which the police responded is 693.7.



EXAMPLE 3-2 Hospital Infections

The data show the number of patients in a sample of six hospitals who acquired an infection while hospitalized. Find the mean.

110 76 29 38 105 31

Source: Pennsylvania Health Care Cost Containment Council.

SOLUTION

$$\bar{X} = \frac{\Sigma X}{n} = \frac{110 + 76 + 29 + 38 + 105 + 31}{6} = \frac{389}{6} = 64.8$$

The mean of the number of hospital infections for the six hospitals is 64.8.

Procedure Table

Finding the Mean for Grouped Data

Step 1 Make a table as shown.

A	B	C	D
Class	Frequency f	Midpoint X_m	$f \cdot X_m$

Step 2 Find the midpoints of each class and place them in column C.

Step 3 Multiply the frequency by the midpoint for each class, and place the product in column D.

Step 4 Find the sum of column D.

Step 5 Divide the sum obtained in column D by the sum of the frequencies obtained in column B.

The formula for the mean is

$$\bar{X} = \frac{\Sigma f \cdot X_m}{n}$$

[Note: The symbols $\Sigma f \cdot X_m$ mean to find the sum of the product of the frequency (f) and the midpoint (X_m) for each class.]

EXAMPLE 3–3 Miles Run per Week

Using the following frequency distribution (taken from Example 2–7), find the mean. The data represent the number of miles run during one week for a sample of 20 runners.

Class boundaries	Frequency
5.5–10.5	1
10.5–15.5	2
15.5–20.5	3
20.5–25.5	5
25.5–30.5	4
30.5–35.5	3
35.5–40.5	2
	<u>20</u>
	Total

SOLUTION

The procedure for finding the mean for grouped data is given here.

Step 1 Make a table as shown.

A Class	B Frequency f	C Midpoint X_m	D $f \cdot X_m$
5.5–10.5	1		
10.5–15.5	2		
15.5–20.5	3		
20.5–25.5	5		
25.5–30.5	4		
30.5–35.5	3		
35.5–40.5	2		
	$n = 20$		

Step 2 Find the midpoints of each class and enter them in column C.

$$X_m = \frac{5.5 + 10.5}{2} = 8 \quad \frac{10.5 + 15.5}{2} = 13 \quad \text{etc.}$$

Step 3 For each class, multiply the frequency by the midpoint, as shown, and place the product in column D.

$$1 \cdot 8 = 8 \quad 2 \cdot 13 = 26 \quad \text{etc.}$$

The completed table is shown here.

A Class	B Frequency f	C Midpoint X_m	D $f \cdot X_m$
5.5–10.5	1	8	8
10.5–15.5	2	13	26
15.5–20.5	3	18	54
20.5–25.5	5	23	115
25.5–30.5	4	28	112
30.5–35.5	3	33	99
35.5–40.5	2	38	76
	$n = 20$		$\Sigma f \cdot X_m = 490$

Find the sum of column D.

Divide the sum by n to get the mean.

$$\bar{X} = \frac{\Sigma f \cdot X_m}{n} = \frac{490}{20} = 24.5 \text{ miles}$$

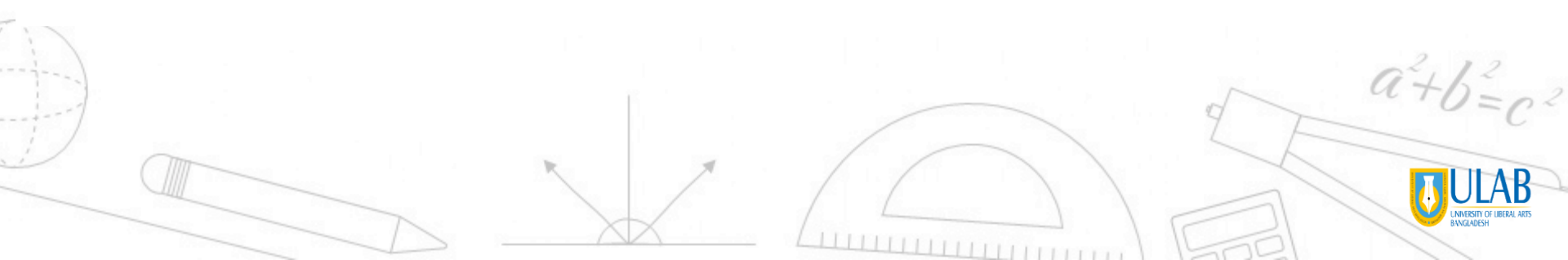


The Median

The **median** is the midpoint of the data array. The symbol for the median is MD.

Procedure Table

Finding the Median

- Step 1** Arrange the data values in ascending order.
- Step 2** Determine the number of values in the data set.
- Step 3**
- a. If n is odd, select the middle data value as the median.
 - b. If n is even, find the mean of the two middle values. That is, add them and divide the sum by 2.
- 

EXAMPLE 3-4 Police Officers Killed

The number of police officers killed in the line of duty over the last 11 years is shown. Find the median.

177 153 122 141 189 155 162 165 149 157 240

Source: National Law Enforcement Officers Memorial Fund.

SOLUTION

Step 1 Arrange the data in ascending order.

122, 141, 149, 153, 155, 157, 162, 165, 177, 189, 240

Step 2 There are an odd number of data values, namely, 11.

Step 3 Select the middle data value.

122, 141, 149, 153, 155, 157, 162, 165, 177, 189, 240



Median

The median number of police officers killed for the 11-year period is 157.

EXAMPLE 3-5 Tornadoes in the United States

The number of tornadoes that have occurred in the United States over an 8-year period follows. Find the median.

684, 764, 656, 702, 856, 1133, 1132, 1303

Source: *The Universal Almanac*.

SOLUTION

Step 1 Arrange the data values in ascending order.

656, 684, 702, 764, 856, 1132, 1133, 1303

Step 2 There are an even number of data values, namely, 8.

Step 3 The middle two data values are 764 and 856.

656, 684, 702, 764, 856, 1132, 1133, 1303

↑

Median

Since the middle point falls halfway between 764 and 856, find the median MD by adding the two values and dividing by 2.

$$\text{MD} = \frac{764 + 856}{2} = \frac{1620}{2} = 810$$

The median number of tornadoes is 810.

Median for Grouped Data

- This is not in your Book
- Please see:

<https://www.youtube.com/watch?v=8tPXYlvbRxA>

Calculate the median from grouped data.

Class interval	f	<cf
84 – 86	2	30
81 – 83	4	28
78 – 80	3	24
75 – 77	5	21
72 – 74	5	16
69 – 71	4	11
66 – 68	3	7
63 – 65	3	4
60 – 62	1	1
N = 30		

$$\tilde{x} = L + \left(\frac{\frac{n}{2} - F_2}{f_2} \right) i$$

Where: L=exact lower limit of the class interval where the median class lies

$\frac{n}{2}$ = median class; i=class interval;

f_2 =frequency of median class

F_2 =cumulative frequency up to the class immediately preceding the median class

$$L = 71.5$$

$$\frac{n}{2} = 15$$

$$i = 3$$

$$F_2 = 11$$

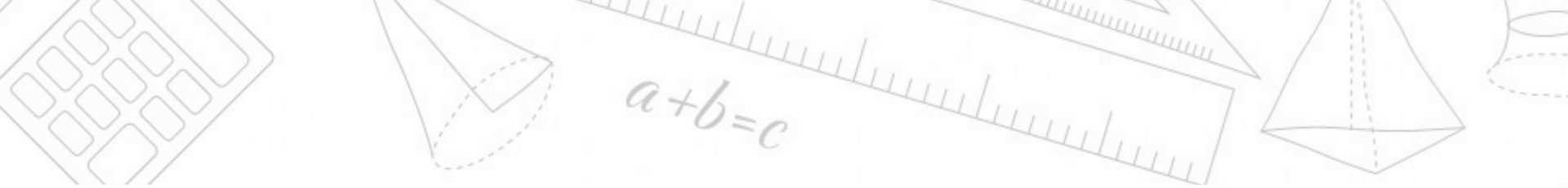
$$f_2 = 5$$

The Mode

- The third measure of average is called the *mode*. The mode is the value that occurs most
- often in the data set. It is sometimes said to be the most typical case.
- The value that occurs most often in a data set is called the **mode**.

The Mode

- A data set that has only one value that occurs with the greatest frequency is said to be **unimodal**.
- If a data set has two values that occur with the same greatest frequency, both values are considered to be the mode and the data set is said to be **bimodal**.
- If a data set has more than two values that occur with the same greatest frequency, each value is used as the mode, and the data set is said to be **multimodal**. When no data value occurs more than once, the data set is said to have *no mode*.
- *Note: Do not say that the mode is zero.* That would be incorrect, because in some data, such as temperature, zero can be an actual value. A data set can have more than one mode or no mode at all. These situations will be shown in some of the examples that follow.



EXAMPLE 3–6 NFL Signing Bonuses

Find the mode of the signing bonuses of eight NFL players for a specific year. The bonuses in millions of dollars are

18.0, 14.0, 34.5, 10, 11.3, 10, 12.4, 10

Source: *USA TODAY*.

SOLUTION

It is helpful to arrange the data in order, although it is not necessary.

10, 10, 10, 11.3, 12.4, 14.0, 18.0, 34.5

Since \$10 million occurred 3 times—a frequency larger than any other number—the mode is \$10 million.

EXAMPLE 3-7 Licensed Nuclear Reactors

The data show the number of licensed nuclear reactors in the United States for a recent 15-year period. Find the mode.

Source: *The World Almanac and Book of Facts*.

104	104	104	104	104
107	109	109	109	110
109	111	112	111	109

SOLUTION

Since the values 104 and 109 both occur 5 times, the modes are 104 and 109. The data set is said to be bimodal.

EXAMPLE 3-8 Accidental Firearm Deaths

The number of accidental deaths due to firearms for a six-year period is shown. Find the mode.

649, 789, 642, 613, 610, 600

Source: National Safety Council.

SOLUTION

Since each value occurs only once, there is no mode.

EXAMPLE 3–9 Miles Run per Week

Find the modal class for the frequency distribution of miles that 20 runners ran in one week, used in Example 2–7.

Class	Frequency
5.5–10.5	1
10.5–15.5	2
15.5–20.5	3
20.5–25.5	5 ← Modal class
25.5–30.5	4
30.5–35.5	3
35.5–40.5	2

SOLUTION

The modal class is 20.5–25.5, since it has the largest frequency. Sometimes the midpoint of the class is used rather than the boundaries; hence, the mode could also be given as 23 miles per week.

EXAMPLE 3-10 Area Boat Registrations

The data show the number of boats registered for six counties in southwestern Pennsylvania. Find the mode.

Westmoreland	11,008
Butler	9,002
Washington	6,843
Beaver	6,367
Fayette	4,208
Armstrong	3,782

Source: Pennsylvania Fish and Boat Commission.

SOLUTION

Since the category with the highest frequency is Westmoreland, the most typical case is Westmoreland. Hence, the mode is Westmoreland.

Exact Mode of Grouped Data

- This is not in your book

- Please Watch:

<https://www.youtube.com/watch?v=Sqkhzdn1eOU>

EXAMPLE 3–11 Salaries of Personnel

A small company consists of the owner, the manager, the salesperson, and two technicians, all of whose annual salaries are listed here. (Assume that this is the entire population.)

Staff	Salary
Owner	\$100,000
Manager	40,000
Salesperson	24,000
Technician	18,000
Technician	18,000

Find the mean, median, and mode.

SOLUTION

$$\mu = \frac{\sum X}{N} = \frac{\$100,000 + 40,000 + 24,000 + 18,000 + 18,000}{5} = \frac{\$200,000}{5} = \$40,000$$

Hence, the mean is \$40,000, the median is \$24,000, and the mode is \$18,000.

Weighted Mean

Sometimes, you must find the mean of a data set in which not all values are equally represented. Consider the case of finding the average cost of a gallon of gasoline for three taxis. Suppose the drivers buy gasoline at three different service stations at a cost of \$3.22, \$3.53, and \$3.63 per gallon. You might try to find the average by using the formula

$$\begin{aligned}\bar{X} &= \frac{\sum X}{n} \\ &= \frac{3.22 + 3.53 + 3.63}{3} = \frac{10.38}{3} = \$3.46\end{aligned}$$

But not all drivers purchased the same number of gallons. Hence, to find the true average cost per gallon, you must take into consideration the number of gallons each driver purchased.

The type of mean that considers an additional factor is called the *weighted mean*, and it is used when the values are not all equally represented.

Find the **weighted mean** of a variable X by multiplying each value by its corresponding weight and dividing the sum of the products by the sum of the weights.

$$\bar{X} = \frac{w_1X_1 + w_2X_2 + \cdots + w_nX_n}{w_1 + w_2 + \cdots + w_n} = \frac{\sum wX}{\sum w}$$

where $w_1, w_2, \dots, w_n$ are the weights and $X_1, X_2, \dots, X_n$ are the values.

Example 3–14 shows how the weighted mean is used to compute a grade point average. Since courses vary in their credit value, the number of credits must be used as weights.

EXAMPLE 3–14 Grade Point Average

A student received an A in English Composition I (3 credits), a C in Introduction to Psychology (3 credits), a B in Biology I (4 credits), and a D in Physical Education (2 credits). Assuming A = 4 grade points, B = 3 grade points, C = 2 grade points, D = 1 grade point, and F = 0 grade points, find the student's grade point average.

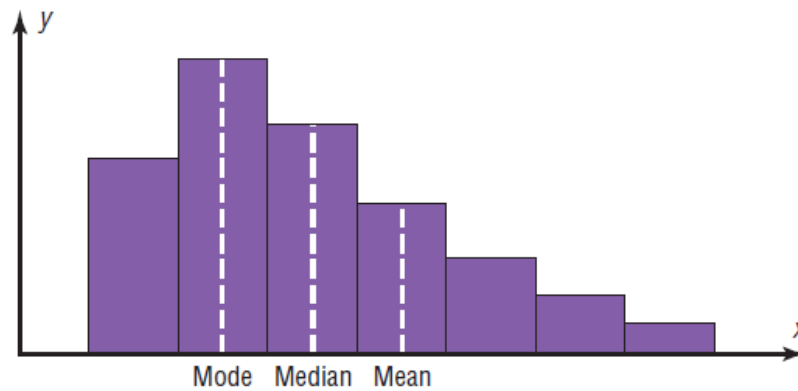
SOLUTION

Course	Credits (w)	Grade (X)
English Composition I	3	A (4 points)
Introduction to Psychology	3	C (2 points)
Biology I	4	B (3 points)
Physical Education	2	D (1 point)

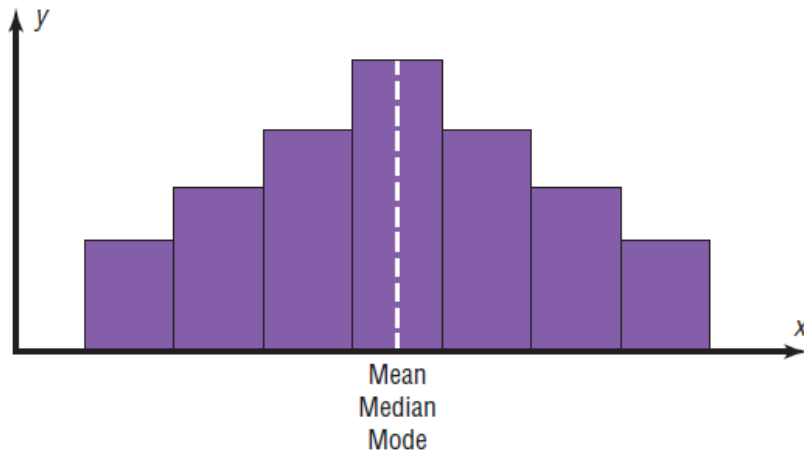
$$\bar{X} = \frac{\sum wX}{\sum w} = \frac{3 \cdot 4 + 3 \cdot 2 + 4 \cdot 3 + 2 \cdot 1}{3 + 3 + 4 + 2} = \frac{32}{12} \approx 2.7$$

The grade point average is 2.7.

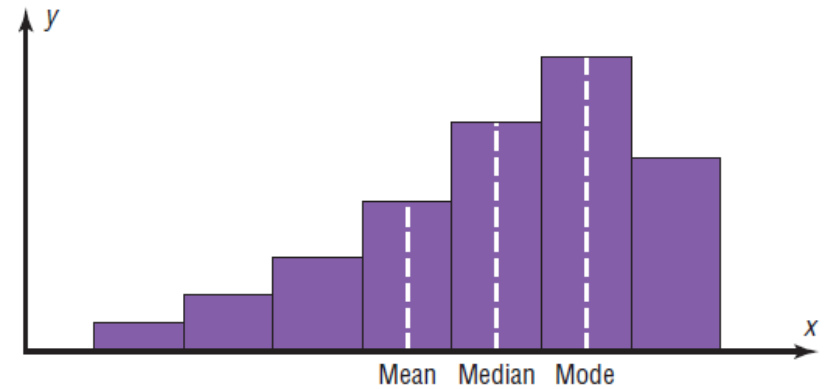
FIGURE 3-1
Types of Distributions



(a) Positively skewed or right-skewed



(b) Symmetric



(c) Negatively skewed or left-skewed

EXAMPLE 3-15 Comparison of Outdoor Paint

A testing lab wishes to test two experimental brands of outdoor paint to see how long each will last before fading. The testing lab makes 6 gallons of each paint to test. Since different chemical agents are added to each group and only six cans are involved, these two groups constitute two small populations. The results (in months) are shown. Find the mean of each group.

Brand A	Brand B
10	35
60	45
50	30
30	35
40	40
20	25

SOLUTION

The mean for brand A is

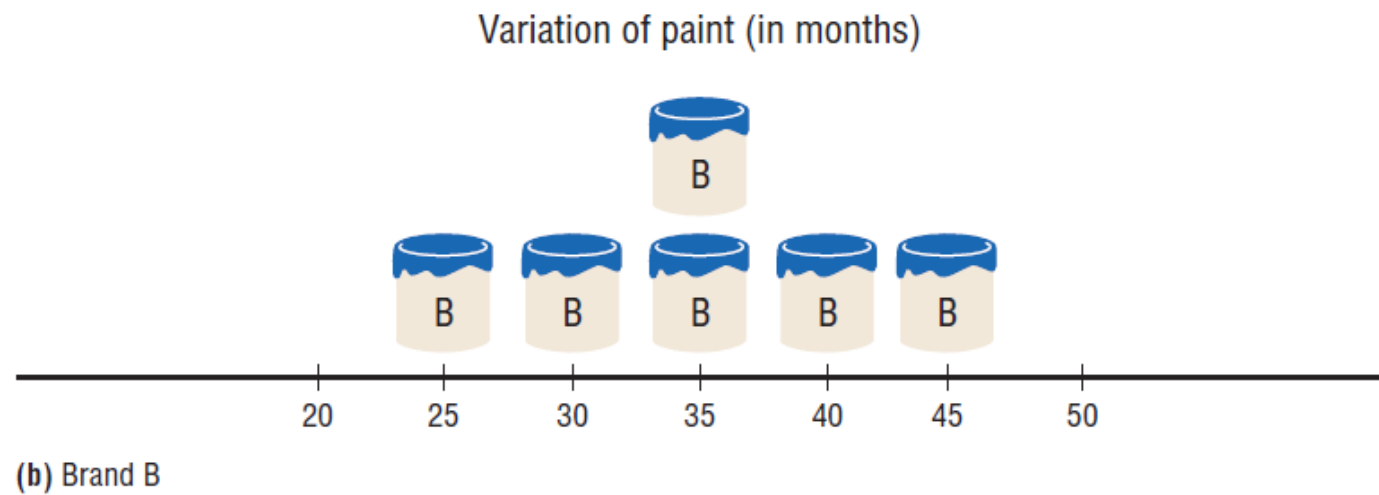
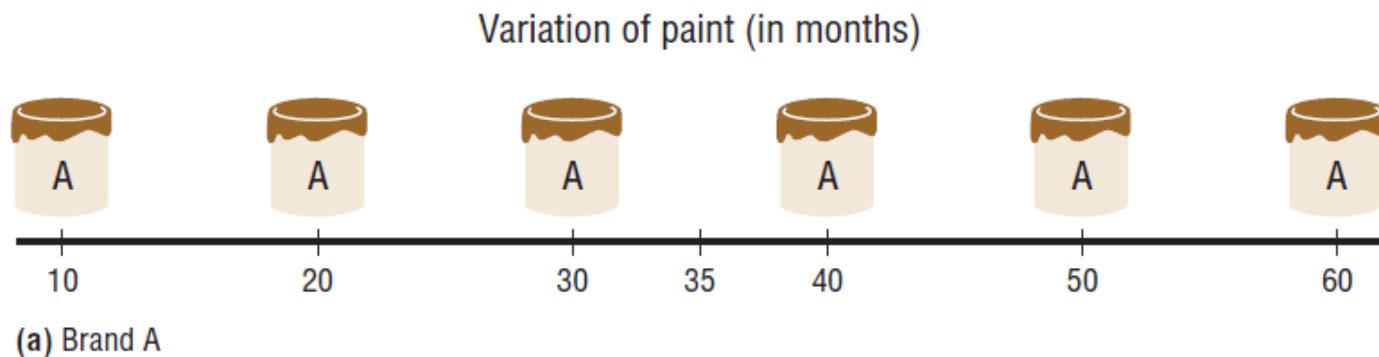
$$\mu = \frac{\Sigma X}{N} = \frac{210}{6} = 35 \text{ months}$$

The mean for brand B is

$$\mu = \frac{\Sigma X}{N} = \frac{210}{6} = 35 \text{ months}$$

Since the means are equal in Example 3–15, you might conclude that both brands of paint last equally well. However, when the data sets are examined graphically, a somewhat different conclusion might be drawn. See Figure 3–2.

As Figure 3–2 shows, even though the means are the same for both brands, the spread, or variation, is quite different. Figure 3–2 shows that brand B performs more



Range

The range is the simplest of the three measures and is defined now.

The **range** is the highest value minus the lowest value. The symbol R is used for the range.

$$R = \text{highest value} - \text{lowest value}$$

EXAMPLE 3-16 Comparison of Outdoor Paint

Find the ranges for the paints in Example 3-15.

SOLUTION

For brand A, the range is

$$R = 60 - 10 = 50 \text{ months}$$

For brand B, the range is

$$R = 45 - 25 = 20 \text{ months}$$

Make sure the range is given as a single number.

The range for brand A shows that 50 months separate the largest data value from the smallest data value. For brand B, 20 months separate the largest data value from the smallest data value, which is less than one-half of brand A's range.

The **population variance** is the average of the squares of the distance each value is from the mean. The symbol for the population variance is σ^2 (σ is the Greek lower-case letter sigma).

The formula for the population variance is

$$\sigma^2 = \frac{\sum(X - \mu)^2}{N}$$

where X = individual value

μ = population mean

N = population size

The **population standard deviation** is the square root of the variance. The symbol for the population standard deviation is σ .

The corresponding formula for the population standard deviation is

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{\sum(X - \mu)^2}{N}}$$

EXAMPLE 3-18 Comparison of Outdoor Paint

Find the variance and standard deviation for the data set for brand A paint in Example 3-15. The number of months brand A lasted before fading was

10, 60, 50, 30, 40, 20

SOLUTION

Step 1 Find the mean for the data.

$$\mu = \frac{\sum X}{N} = \frac{10 + 60 + 50 + 30 + 40 + 20}{6} = \frac{210}{6} = 35$$

Step 2 Subtract the mean from each data value ($X - \mu$).

$$\begin{array}{lll} 10 - 35 = -25 & 50 - 35 = +15 & 40 - 35 = +5 \\ 60 - 35 = +25 & 30 - 35 = -5 & 20 - 35 = -15 \end{array}$$

Step 3 Square each result $(\bar{X} - \mu)^2$.

$$\begin{array}{lll} (-25)^2 = 625 & (+15)^2 = 225 & (+5)^2 = 25 \\ (+25)^2 = 625 & (-5)^2 = 25 & (-15)^2 = 225 \end{array}$$

Step 4 Find the sum of the squares $\sum(\bar{X} - \mu)^2$.

$$625 + 625 + 225 + 25 + 25 + 225 = 1750$$

Step 5 Divide the sum by N to get the variance $\frac{[\sum(\bar{X} - \mu)^2]}{N}$.

$$\text{Variance} = 1750 \div 6 \approx 291.7$$

Step 6 Take the square root of the variance to get the standard deviation. Hence, the standard deviation equals $\sqrt{291.7}$, or 17.1. It is helpful to make a table.

Column A contains the raw data X . Column B contains the differences $X - \mu$ obtained in step 2. Column C contains the squares of the differences obtained in step 3.

A Values X	B $X - \mu$	C $(X - \mu)^2$
10	-25	625
60	+25	625
50	+15	225
30	-5	25
40	+5	25
20	-15	225
		1750

EXAMPLE 3-19 Comparison of Outdoor Paint

Find the variance and standard deviation for brand B paint data in Example 3-15. The months brand B lasted before fading were

35, 45, 30, 35, 40, 25

SOLUTION

Step 1 Find the mean.

$$\mu = \frac{\Sigma X}{N} = \frac{35 + 45 + 30 + 35 + 40 + 25}{6} = \frac{210}{6} = 35$$

Step 2 Subtract the mean from each value, and place the result in column B of the table.

$$\begin{array}{lll} 35 - 35 = 0 & 45 - 35 = 10 & 30 - 35 = -5 \\ 35 - 35 = 0 & 40 - 35 = 5 & 25 - 35 = -10 \end{array}$$

Step 3 Square each result and place the squares in column C of the table.

Step 4 Find the sum of the squares in column C.

$$\Sigma(X - \mu)^2 = 0 + 100 + 25 + 0 + 25 + 100 = 250$$

Step 5 Divide the sum by N to get the variance.

$$\sigma^2 = \frac{\Sigma(X - \mu)^2}{N} = \frac{250}{6} = 41.7$$

Step 6 Take the square root to get the standard deviation.

$$\sigma = \sqrt{\frac{\Sigma(X - \mu)^2}{N}} = \sqrt{41.7} \approx 6.5$$

Hence, the standard deviation is 6.5.

A X	B $X - \mu$	C $(X - \mu)^2$
35	0	0
45	10	100
30	-5	25
35	0	0
40	5	25
25	-10	100

Formula for Sample Variance & Standard Deviation

Formula for the Sample Variance

The formula for the sample variance (denoted by s^2) is

$$s^2 = \frac{\Sigma(X - \bar{X})^2}{n - 1}$$

where X = individual value

$\bar{X}$ = sample mean

n = sample size

Formula for the Sample Standard Deviation

The formula for the sample standard deviation, denoted by s , is

$$s = \sqrt{s^2} = \sqrt{\frac{\Sigma(X - \bar{X})^2}{n - 1}}$$

where X = individual value

$\bar{X}$ = sample mean

n = sample size



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Thank You

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